

Employing Metaprogramming: A Macro-Driven Strategy to Overcome Compiler-Imposed Limitations on Specialization

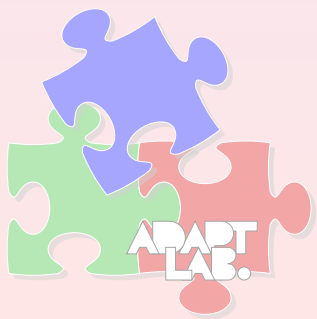
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Computer Science Department
BSc in Computer Science

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Co-Advisor: M.Sc. Federico Bruzzone

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L-31 - Computer science
Academic Year 2024-2025





State of the Art

State of the Art

Limitations of Current Approaches

Problem Statement

Proposal

By Example: Implementing the Proposal

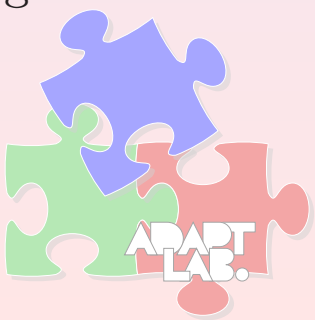
Limitations and Future Work

Implementation Details

Evaluation

Conclusions

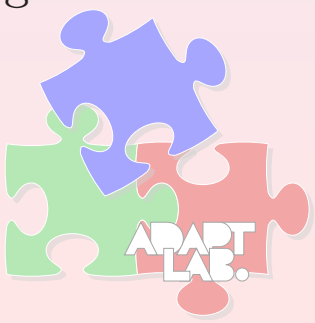
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Limitations of Current Approaches

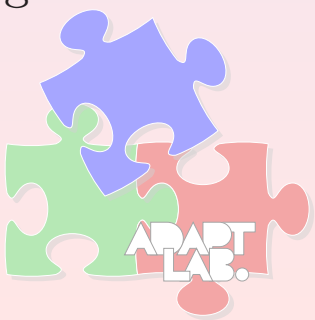
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Problem Statement

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Problem Statement
Motivational Example

State of the Art

Limitations of Current Approaches

Problem Statement

Proposal

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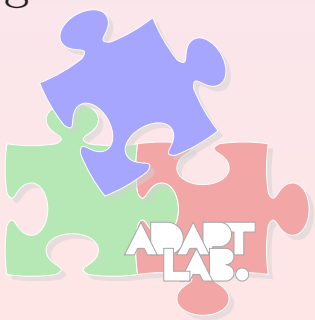
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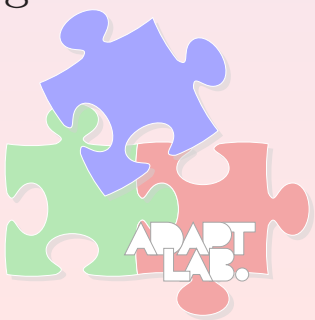
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Proposal

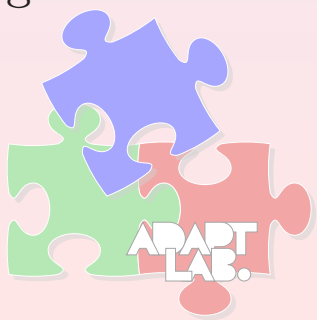
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By Example: Implementing the Proposal

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Limitations and Future Work

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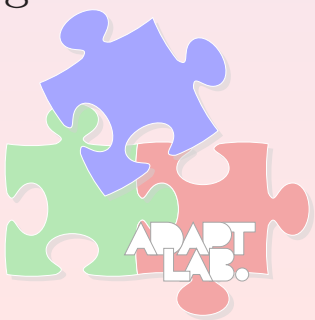
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Implementation Details
Conditions Definition

State of the Art

Limitations of Current Approaches

Problem Statement

Proposal

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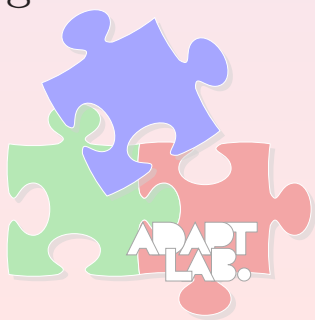
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Implementation Details
Method Wrapping

State of the Art

Limitations of Current Approaches

Problem Statement

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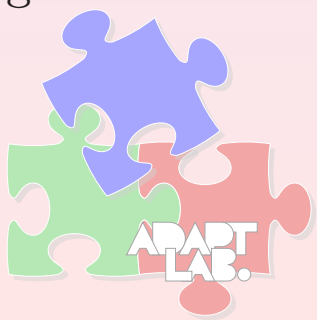
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Implementation Details
Build Scripts

State of the Art

Limitations of Current Approaches

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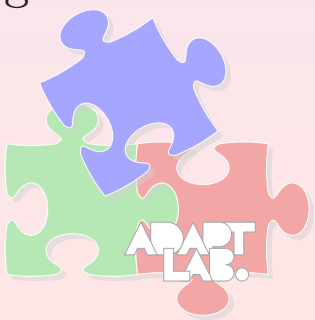
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Evaluation
Compiler Instrumentation

State of the Art

Limitations of Current Approaches

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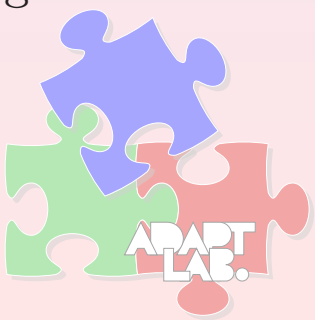
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Evaluation
Specialization Heuristics

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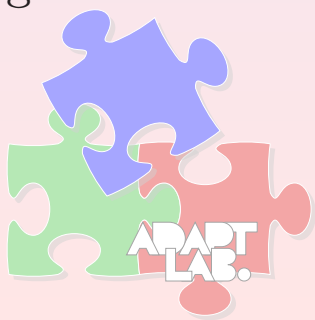
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Evaluation
Results

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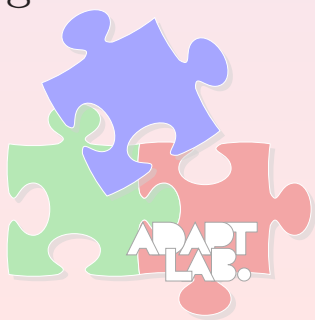
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Conclusions

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